AUTO-CONFIGURATION OF IP ADDRESSES IN MOBILE AD-HOC NETWORKS (MANET)

Summary

Objectives of this study were to examine and evaluate the existing IP address auto-configuration and management methods in Mobile Ad hoc Area Networks and further to propose architecture for the IP address auto-configuration and management.

Chapter-I describes the concept of mobile ad hoc network and highlights various issues that need immediate attention to increase the use of MANET. In this chapter objective of study, research methodology and major contributions of this thesis are also presented.

Chapter-II describes the challenges in IP address auto-configuration problem and also the trends in development of these challenges. To find out these trends, electronic resources such as IEEE Xplore™ and Springer's Lecture Notes in Computer Science are used. It is found that approximately only 5% of the total research publications in MANET area are there in the literature that relates to the IP address auto-configuration problem.

Chapter-III elaborates the analysis of existing popular address auto-configuration protocols. A basic characteristics and complexity comparison in terms of memory overhead, latency and communication overhead have also been presented.

Focus of the Chapter-IV is on simulated performance analysis. Network simulator (ns-2, version 2.29) has been used to perform these experiments using various simulation parameters and mobility patterns like obstacle mobility model, random way point mobility model. In first set of experiments

it has been found that DSDV routing protocol is performing better as compared to TORA, DSR and AODV. Based on these findings, DSDV has been used as routing protocol to conduct simulated experiments for eight address auto-configuration protocols. Prophet, DPDAA, PACMAN and DHAPM protocols have been analyzed using obstacle mobility model. MANETConf, Prime DHCP, ZAL and CFAA have been analyzed using random way point mobility model. The simulation experiments are conducted for a medium size campus with obstacles, in which there are at most 80 nodes participating to form the network, using Network Simulator (NS-2). Area of simulation is of size 600m x 1000m and a node density of 80 in varying mobility, flows and network load. It is found that PACMAN is performing suitably as compared Prophet, DPDAA and DHAPM in two factors but address assignment latency in this case is high.

In Chapter-V, a new algorithm and architecture namely Extended Prime Number based Allocation (EPNA), have been proposed for IP address allocation. It has also been shown that EPNA gives bounded delay and communication overhead for IP address assignment to new nodes. Increase in average values of overhead and delay is also very slow. It almost remains constant and with in bounded limits. A simulation program has been designed in ‘c’ to run various experiments to verify the proposed technique and validate the analytical model. Results of the various experiments using this simulation program shows that EPNA gives bounded delay and latency for various loss probabilities. Effect of address space range on the latency and overhead in address assignment is also studied using this simulation program. Comparison of PNAA and EPNA shows that EPNA performs better than PNAA.